

THE ILLUSION OF FINANCIAL CONTROL

Literacy, Overconfidence and Financial Vulnerability in Italy

PREPARED BY

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Bank of Italy IACOFI Survey Wave 2023

Primary Dataset

Analysis Period: May 2026

1. Abstract

This study investigates whether financial capability is sufficient to protect Italian adults from financial vulnerability. The project starts from a practical policy problem: people may possess financial knowledge, feel confident, and use digital financial tools, yet still experience difficulty dealing with financial shocks. Using the 2023 Bank of Italy IACOFI microdata, financial capability is measured through objective financial knowledge, self-assessed knowledge, and digital financial engagement. Financial vulnerability is measured through difficulty making ends meet, ability to face an unexpected expense, and capacity to survive after loss of income. The analysis was conducted in R through data cleaning, construction of composite indicators, correlation analysis, linear regression models, and an overconfidence/miscalibration test. The results show that financial capability, as a broad construct, is not significantly associated with lower vulnerability once age, income, and gender are controlled for. Income is consistently associated with lower vulnerability, confirming the importance of material resources. An interaction model shows that the association between capability and vulnerability does not substantially differ across income levels. The most important finding is that overconfidence, measured as a mismatch between perceived and actual knowledge, is positively and significantly associated with financial vulnerability. The study therefore concludes that financial capability alone is not enough: accurate self-assessment also matters.

2. Research question

2a. Statement

In modern financial systems, individuals are increasingly expected to manage their own financial well-being. Access to financial information has expanded, digital tools have become widespread, and financial literacy programmes have been promoted as a key solution to financial fragility. As a result, many individuals appear financially capable: they understand basic financial concepts, use digital financial services, and feel confident in their financial decisions.

However, a critical question remains: does this apparent capability actually translate into financial security? Or is it possible that individuals who appear financially capable remain vulnerable in practice? This project investigates this tension by examining whether financial capability truly protects individuals, or whether a gap exists between perceived control and actual financial resilience.

Central Research Question:

Is financial capability sufficient to protect individuals from financial vulnerability, or does vulnerability persist even among those who appear financially capable?

The empirical analysis is organized around three supporting questions:

- Is broad financial capability, including literacy, self-assessed knowledge and digital engagement, associated with lower financial vulnerability?
- Does the association between financial capability and vulnerability depend on income level?
- Is overconfidence, measured as a mismatch between perceived and actual financial knowledge, associated with higher financial vulnerability?

2b. Motivation and Stakeholders/Users

The motivation for the study is that financial education is often presented as a solution to household financial fragility. This assumption is reasonable but incomplete. A person can understand interest rates or inflation and still be unable to pay an unexpected bill. Similarly, access to digital tools may make financial management easier, but it does not automatically create savings or remove economic constraints. The project therefore focuses on the gap between apparent financial control and actual financial resilience.

This topic is relevant for public institutions and policymakers because financial education programmes should not only increase knowledge; they should also help individuals judge their own limits correctly. The Bank of Italy, Schools, Universities, Consumer protection agencies, Financial Institutions, and Digital banking providers may all benefit from understanding whether knowledge and digital engagement are enough to reduce vulnerability. The study is also relevant for households, because it highlights that feeling capable is not the same as being financially secure.

The contribution of the project is not to claim a causal effect. Rather, it provides a structured empirical assessment of whether capability and vulnerability move together in the expected way. If they do not, the project asks what this says about the limits of financial literacy and the possible role of overconfidence. This is consistent with the course objective: starting from a good question, using the available data realistically, and interpreting results carefully.

3. The Data

3a. Data Description

The analysis is based on the 2023 wave of the Italian Survey on Financial Literacy and Digital Financial Skills in Italy: Adults, known as IACOFI, conducted by the Bank of Italy [7]. The survey is based on the OECD/INFE methodology for measuring financial literacy and financial inclusion [4]. The Bank of Italy provides anonymized raw microdata for research use in several formats, including CSV, SAS and Stata. The CSV version was used for this project because it can be imported directly into R and supports full reproducibility of the analysis.

The raw dataset contains 4,862 observations and 219 variables. Each observation represents an adult respondent. The variables cover personal and household characteristics, financial planning and management, active saving and financial shocks, financial products and services, digital financial activity, objective financial knowledge, attitudes, behaviour and background information. The data description file was essential because many variables are coded with short names such as QD7, QF4 or QK1. Without the documentation, these codes would not be interpretable.

The variables used in this project were selected because they correspond directly to the research question. Demographic controls include gender, age and income. Gender is recorded as QD1, age as QD7, and income band as QD13. Financial vulnerability is represented by QF11, QF4 and QF13. QF11 records whether the respondent's income failed to cover living expenses in the previous 12 months. QF4 records whether the respondent could pay a major expense equivalent to monthly income without borrowing or asking others for help. QF13 records how long the respondent could cover living expenses after losing the main source of income.

Financial capability is represented by three dimensions. First, objective financial knowledge is measured through questions on inflation, simple interest and compound interest. Second, perceived knowledge is measured through QK1, where respondents rate their own knowledge of financial matters compared with others. Third, digital engagement is measured through QP8 variables, which capture whether respondents have carried out online financial activities, such as opening an account online, requesting a card online, subscribing to insurance online, taking credit online, or investing through online platforms.

The workflow began by renaming the most relevant variables using the data description. The full dataset was then reduced from 219 variables to 28 variables directly related to the study. After cleaning non-response codes and assessing missingness, variables in the QK7 block were removed because they had substantial missingness and were not necessary for the construction of the final indicators. The final analytical dataset retained 4,862 observations and 22 selected variables before construction of the indices. This reduction was not done for convenience alone; it was done to keep the analysis focused on the research question and avoid unnecessary complexity.

3b. Strengths and Weaknesses of the Data with respect to the Study Goal

Strengths.

- ❖ The main strength of the dataset is its multidimensional structure. It does not measure only knowledge. It also includes perceived knowledge, financial behaviour, vulnerability indicators, and digital financial activities. This is particularly useful for this project because the research question is about the possible mismatch between appearing capable and actually being financially secure.
- ❖ A second strength is the presence of both objective and subjective measures. Objective knowledge questions allow the construction of a financial knowledge score, while self-assessed knowledge allows the study to approximate overconfidence or

- ❖ miscalibration. This distinction is central to the project because an illusion of control can only be studied if perceived ability and actual ability can be compared.
- ❖ A third strength is that financial vulnerability is measured directly through practical questions. The data do not only ask whether respondents know financial concepts. They also ask whether respondents can face an unexpected expense, whether income has failed to cover living expenses, and how long they could survive without income. These are concrete indicators of fragility and resilience.
- ❖ A fourth strength is the inclusion of digital financial engagement. The 2023 wave is particularly useful because it includes questions related to digital financial skills and online financial behaviour. This allows the study to reflect a current real-life issue: the possibility that using financial technology may create a sense of control without necessarily guaranteeing financial resilience.

Weaknesses and limitations.

- ❖ The most important limitation is that the dataset is cross-sectional. It provides a snapshot of individuals at one point in time. Therefore, the analysis can identify associations but cannot prove causality. For example, if overconfidence is associated with higher vulnerability, the study cannot prove that overconfidence caused vulnerability. It can only say that they are statistically associated in the data.
- ❖ A second limitation is that several variables are self-reported. Respondents may misremember, refuse to answer, or give socially desirable responses. This is especially relevant for questions about income, financial difficulty and self-assessed knowledge. The project treats these limitations carefully and does not overstate the results.
- ❖ A third limitation is missing data. The dataset contains special codes for 'do not know', 'refused', 'not applicable' and irrelevant answers. These were converted into missing values before analysis. Some variables still had considerable missingness after cleaning. Instead of deleting observations blindly, the project examined missingness across variables and removed only variables that were both highly incomplete and not essential for the research question.
- ❖ A fourth limitation is that the concept of illusion of financial control is not directly observed. The dataset does not ask respondents whether they feel falsely protected. The study therefore uses proxies: broad financial capability, self-assessed knowledge, and the gap between perceived and actual knowledge. This is acceptable for the present project, but it means that results must be interpreted as evidence of a possible mismatch rather than direct proof of a psychological illusion.
- ❖ Finally, many variables are categorical or ordinal. This limits precision. For example, income is measured in bands rather than exact monthly income. The analysis therefore treats the results as indicative and policy-relevant rather than exact measurement of financial positions.

4. Methodology

4a. The Overall Logic Thread of the Study

The logic of the study follows the sequence problem -> concept -> measurement -> test -> interpretation.

The problem is that individuals may appear financially capable but still be vulnerable. The concepts are financial capability, financial vulnerability, and overconfidence or miscalibration. The measurement step translates these concepts into variables using the Bank of Italy survey. The testing step uses correlation and regression models to examine whether the data support the idea that capability protects individuals from vulnerability. The interpretation step asks whether the results are consistent with an illusion of financial control.

The analysis deliberately avoids a machine-learning style approach. The aim is not to compare many algorithms or maximize prediction. The aim is to use a small number of interpretable statistical tools to answer a conceptually relevant question. This is why the project uses descriptive statistics, composite indices, correlation, linear regression, an interaction model, and a mismatch analysis based on overconfidence.

- ✓ The first methodological step was to import the raw CSV file into R and inspect the number of rows and columns. This confirmed that the raw dataset contained 4,862 respondents and 219 variables.
- ✓ The second step was to use the codebook to rename important variables. This made the analysis readable and reduced the risk of interpreting coded variables incorrectly.
- ✓ The third step was variable selection. Variables were selected according to the research question rather than technical convenience. The selected variables represented demographic characteristics, financial vulnerability, objective and subjective knowledge, and digital engagement.
- ✓ The fourth step was cleaning. Special response codes such as -97, -98, -99 and -999 were converted into missing values across the selected dataset. This was done globally, not variable by variable, to keep the workflow coherent and reproducible.
- ✓ The fifth step was missingness analysis. The percentage of missing values was computed for each selected variable. Variables from the QK7 block were removed because they showed high missingness and were not essential for the final measurement strategy. Crucially, variables such as income, vulnerability indicators and knowledge variables were retained despite missingness because they are central to the research question.
- ✓ The sixth step was construction of analytical variables. A financial vulnerability index was created from financial difficulty, shock ability and survival capacity. A financial knowledge index was created from correct answers to knowledge questions. A digital engagement index was created from online financial activity variables. A financial capability index was then created by combining financial knowledge, self-assessed knowledge and digital engagement. Finally, an overconfidence or mismatch variable was created by comparing self-assessed knowledge with actual knowledge.
- ✓ The seventh step was analysis. The study first examined the simple correlation between financial capability and financial vulnerability. It then estimated a main regression model controlling for age, income and gender. To check whether capability works differently depending on income, an interaction model between financial capability and income was estimated.

- ✓ Finally, an overconfidence model was estimated to examine whether mismatch between perceived and actual knowledge is associated with vulnerability.

4b. The Tools: Which and Why

The analysis was conducted in R. R was chosen because it is suitable for reproducible data analysis, statistical modelling and report generation. The tidyverse package was used for data import, variable selection, data cleaning and manipulation.

Base R and the `lm()` function were used for correlation and linear regression models. `ggplot2` was used to produce the figure showing the association between overconfidence and financial vulnerability.

Linear regression was used because the financial vulnerability index is treated as a continuous composite indicator. Logistic regression was not used in the final model because the outcome was not coded as a binary yes/no variable. This choice keeps the analysis aligned with the constructed index and allows interpretation of changes in the vulnerability score.

The control variables were included to avoid a naive interpretation. Income is particularly important because financial vulnerability is strongly linked to economic resources. Age and gender were included to account for demographic variation. The interaction model was added because it was possible that capability might matter only at certain income levels. The overconfidence model was added because the title and research question focus on the mismatch between perceived control and actual vulnerability.

5. Results

5a. The Steps of the Study: Description and Essential numbers/pictures

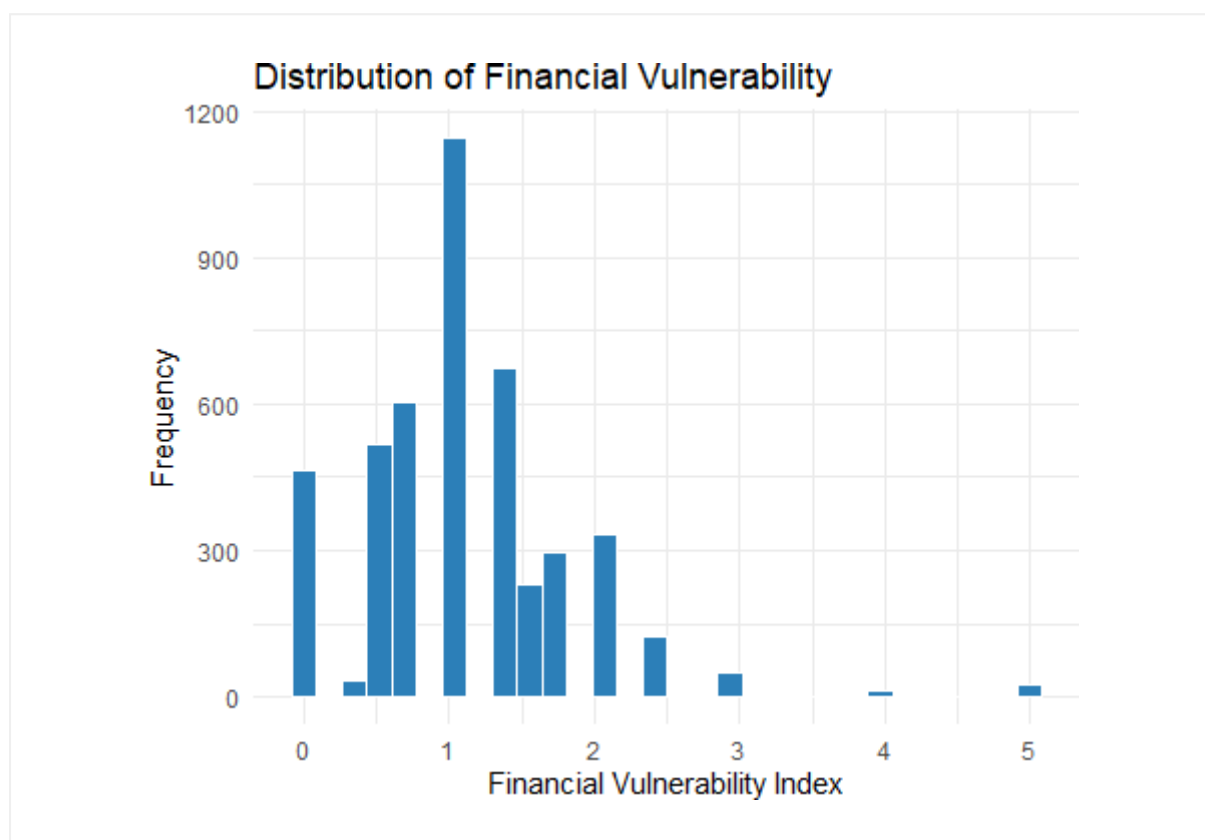


FIGURE A: FINANCIAL VULNERABILITY INDEX

The financial vulnerability index was constructed to summarize financial difficulty, inability or reduced ability to face shocks, and limited survival capacity after loss of income. The financial knowledge index ranged from 0 to 1 and had an average of 0.621, meaning that respondents answered about 62 percent of the selected knowledge questions correctly on average. The digital financial engagement index had a low mean of 0.115, showing that online financial activity was limited for many respondents. The final financial capability index, combining knowledge, self-assessment and digital engagement, had a mean of 0.4675.

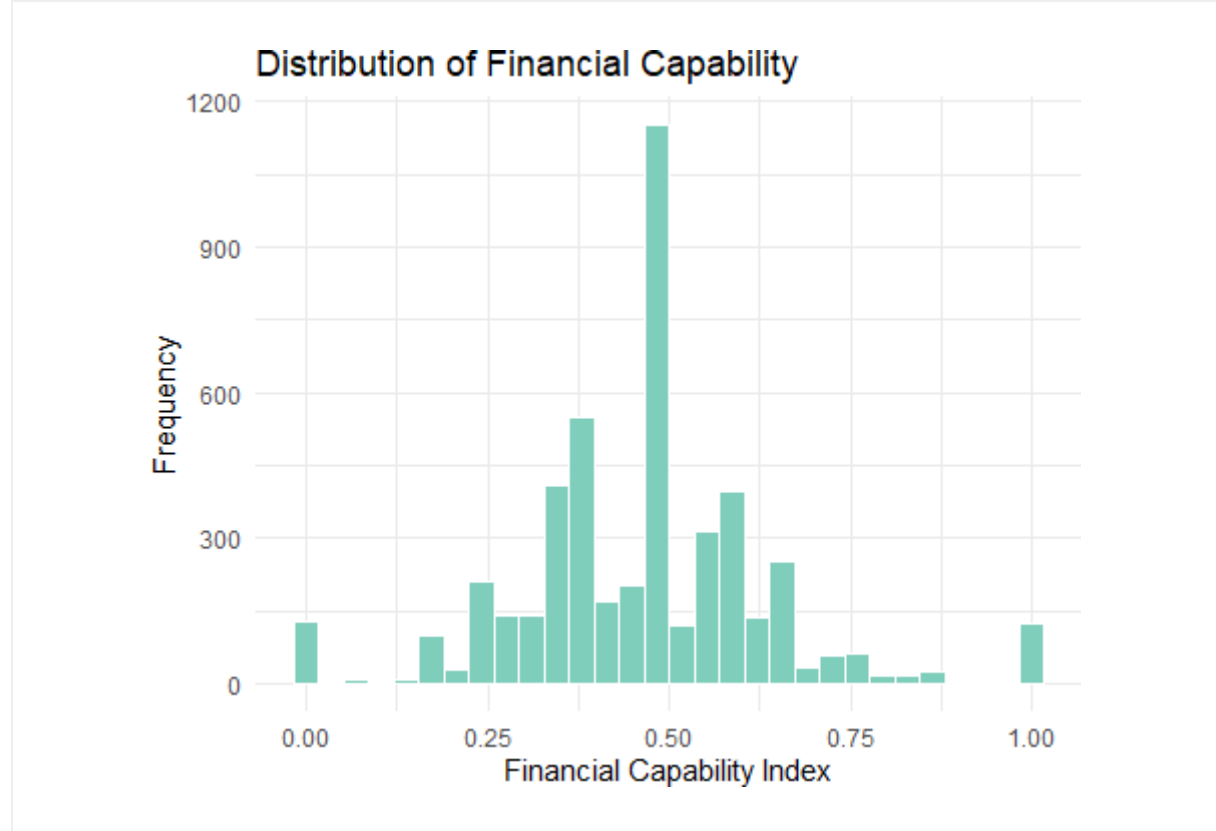


Figure B: Financial Capability Index

Figure B shows that most individuals exhibit moderate levels of financial capability, with values concentrated around the middle of the index. This indicates that financial knowledge, confidence, and digital engagement are relatively widespread. Based on this, one would expect higher capability to be associated with lower financial vulnerability. The following analysis tests whether this expectation is supported by the data.

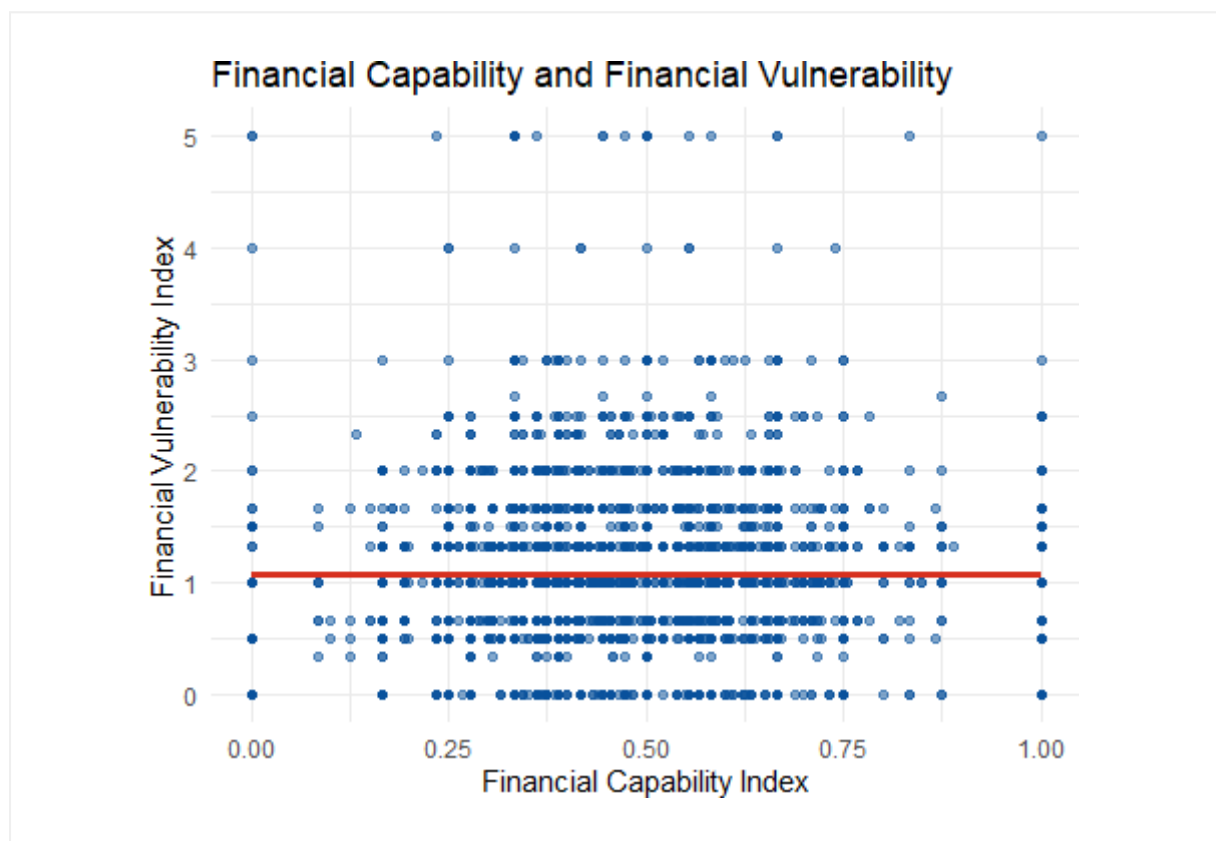


FIGURE C: RELATIONSHIP BETWEEN FINANCIAL CAPABILITY AND FINANCIAL VULNERABILITY.

Figure C presents the relationship between financial capability and financial vulnerability. The correlation between financial capability and financial vulnerability was approximately 0.003. This is close to zero and suggests that there is no meaningful simple linear association between the two indices. In practical terms, individuals with higher capability are not clearly less vulnerable in the raw association. This finding already challenges the simple expectation that capability automatically protects individuals from vulnerability.

The main regression model then tested whether financial capability is associated with vulnerability after controlling for age, income and gender. The coefficient of financial capability was 0.0386 and was not statistically significant ($p = 0.607$). Therefore, the null hypothesis that financial capability has no association with financial vulnerability could not be rejected. Income, in contrast, was negative and highly significant, meaning that higher-income respondents were less vulnerable. Age was also negative and significant in this model, while gender was not significant.

Model / variable	Coefficient	p-value	Interpretation
Main model: Financial capability	0.0386	0.607	Not significant; capability does not significantly reduce vulnerability
Main model: Income	-0.1206	<0.001	Higher income is associated with lower vulnerability
Main model: Age	-0.0038	<0.001	Older respondents are slightly less vulnerable in this model
Main model: Gender	-0.0260	0.283	Not statistically significant
Model R-squared	0.0238	--	Small explanatory power, expected in social survey data

TABLE D1: OUTPUT FROM REGRESSION ANALYSIS

The interaction model asked whether financial capability becomes protective at different income levels. This is important because capability may be useful only when a person has sufficient resources to act on that capability. The interaction between financial capability and income was not statistically significant ($p = 0.717$). This suggests that, within this model, the effect of capability does not meaningfully differ across income groups. Income itself remained relevant, but it did not make financial capability significantly protective.

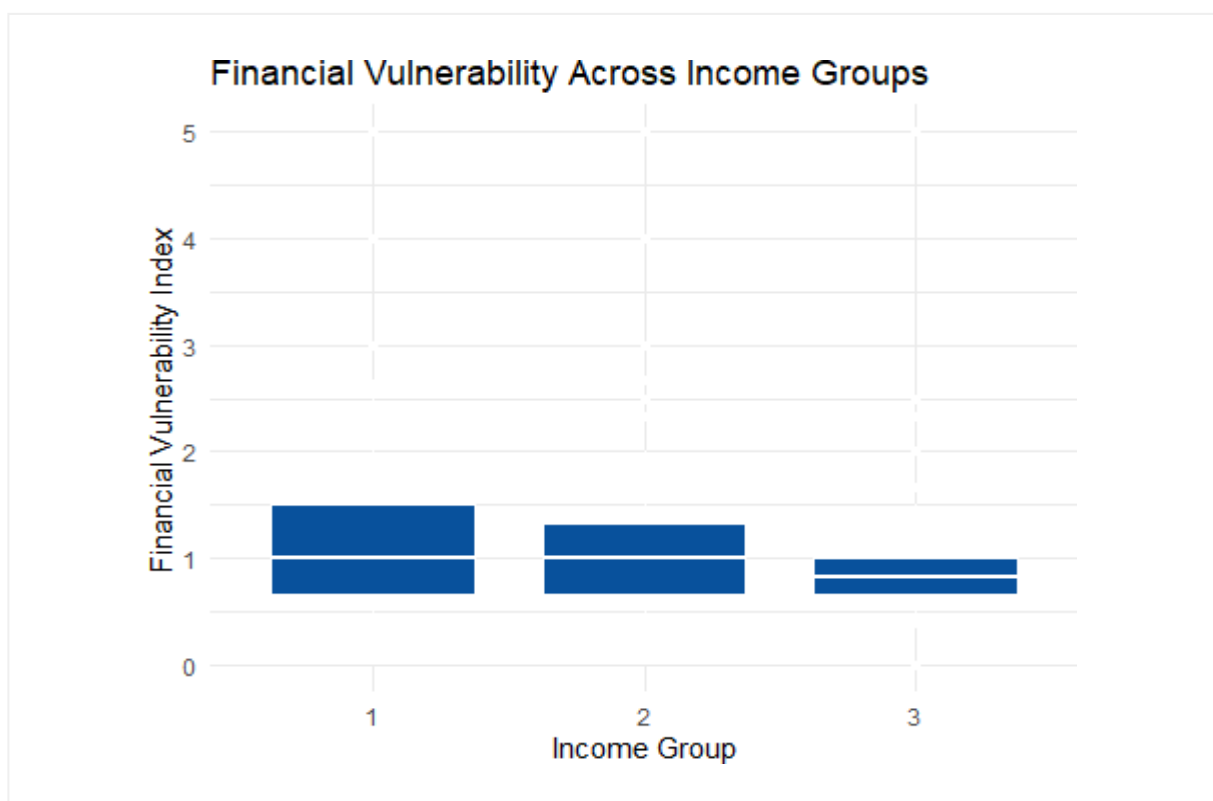


FIGURE D. FINANCIAL VULNERABILITY ACROSS INCOME GROUPS

Figure D shows that financial vulnerability decreases as income increases. Higher-income individuals have lower and more stable vulnerability levels, while lower-income groups show higher and more varied vulnerability. This suggests that income plays an important protective role, and that financial resources may matter more than financial capability alone.

This interaction model examined overconfidence, understood as a mismatch between perceived and actual knowledge. This model directly relates to the title of the project. The coefficient of the overconfidence gap was positive and statistically significant (coefficient = 0.1323, $p < 0.001$). This means that respondents showing greater mismatch between perceived and actual financial knowledge were more financially vulnerable, even after controlling for age, income and gender.

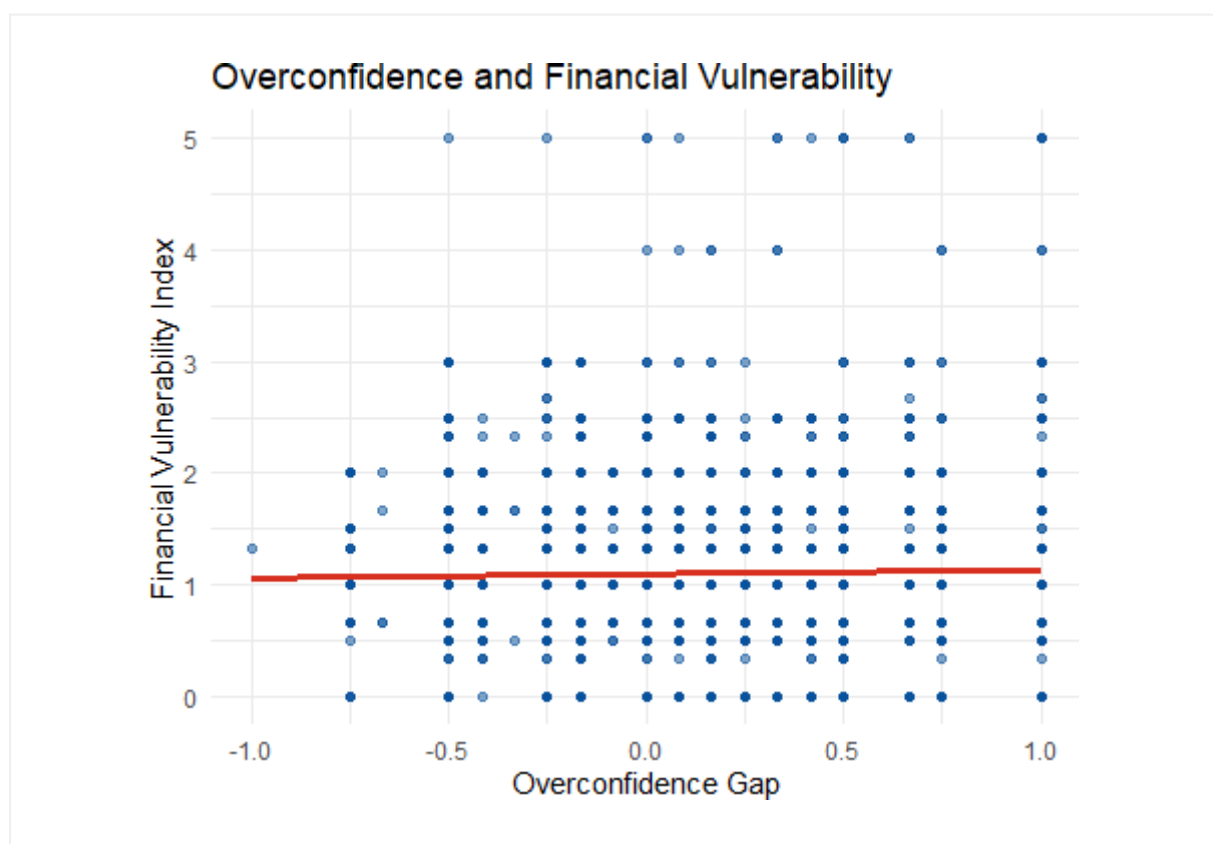


FIGURE E. OVERCONFIDENCE GAP AND FINANCIAL VULNERABILITY

The results provide a clear answer to the research question. Financial capability alone is not enough to protect individuals from financial vulnerability. Both the correlation and regression results show that capability is not a strong predictor of vulnerability.

Income plays a more important role. Higher income is associated with lower financial vulnerability, showing that financial resources are key for financial stability.

The interaction model shows that financial capability does not become more effective at different income levels. This means that capability does not suddenly protect individuals just because income changes.

The most important finding comes from the overconfidence model. Individuals who overestimate their financial knowledge are more likely to be financially vulnerable. This supports the idea of an “illusion of financial control,” where people appear confident but remain financially at risk.

The figure supports this result. Although the points are spread out, the fitted line slopes slightly upward, showing a positive relationship. This suggests that overconfidence contributes to vulnerability, even if it is not the only factor.

Finally, the low R-squared values do not mean the model is weak. Financial vulnerability depends on many other factors not fully captured in the dataset, such as debt, expenses, and exposure to financial shocks. The model is still useful because it answers the main research question.

6. Conclusion

The goal of this study was to examine whether financial capability is sufficient to protect Italian adults from financial vulnerability. The answer from the analysis is: not by itself. Broad financial capability, measured through knowledge, self-assessed knowledge and digital engagement, does not significantly reduce financial vulnerability once demographic and income controls are included. This is the main gap between the expected theoretical goal and the empirical result.

The results suggest that financial capability should not be treated as a complete solution to financial fragility. It is useful, but it is not enough. Income remains important, and overconfidence or miscalibration appears to matter. The study therefore supports a more careful view of financial education: it should not only teach concepts, but also help individuals recognize the limits of their own knowledge and understand when they may be exposed to risk.

The main strength of the study is its conceptually focused question. Instead of asking a generic question such as whether financial literacy improves behaviour, it asks whether apparent financial control actually protects people. The use of Bank of Italy microdata makes the analysis credible and policy-relevant. The construction of capability, vulnerability and overconfidence indicators allows the project to connect the data directly to the research question.

The main weakness is that the analysis is based on cross-sectional data. The results are associations, not causal effects. Another weakness is that the illusion of financial control is not directly measured. It is approximated through self-assessed knowledge, actual knowledge and digital engagement. This is a reasonable approach given the available data. Although the dataset includes proxies for confidence and behavioural bias, future research could use more direct measures of perceived control, risk perception, and behavioural biases.

Future research could improve the analysis in several ways. First, panel data could be used to study whether capability and vulnerability change over time. Second, more precise income, wealth and debt variables would allow a clearer distinction between knowledge and resources. Third, experimental or longitudinal studies could test whether financial education programmes reduce overconfidence or simply increase confidence. Fourth, digital financial behaviour could be studied more deeply by distinguishing between passive use of tools and active, informed financial management.

Overall, the project shows that financial vulnerability is not simply a problem of ignorance. It is also a problem of resources, resilience and self-assessment. The main lesson is that people may appear financially capable and still be vulnerable. For policymakers and educators, this means that financial literacy programmes should be paired with realistic guidance, consumer protection, and tools that help individuals understand not only financial products, but also their own limits.

7. References

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8. Appendix

Appendix A: Descriptive Statistics

Table A1. Summary Statistics of Key Variables

Variable	Mean	Median	Min	Max	Notes
Financial Vulnerability Index	1.07	1.00	0	5	Composite measure
Financial Knowledge Index	0.62	0.67	0	1	Based on correct answers
Digital Engagement Index	0.12	0.00	0	1	Low usage overall
Financial Capability Index	0.47	~0.47	0	1	Combined index
Age	50.3	50	18	79	Years
Income Group	1.69	2	1	3	Categorical

Table A2. Missing Data Summary (Selected Variables)

Variable	Missing (%)
Income	31.1%
Interest Knowledge	29.9%
Survival Capacity	29.6%
Shock Ability	27.5%
Compound Interest Knowledge	22.5%

Appendix B: Variable Construction

Table B1. Construction of Key Indices

Index	Components	Method
Financial Vulnerability	Shock ability, financial difficulty, survival capacity (reversed)	Average
Financial Knowledge	Inflation, interest, compound interest	Binary scoring (correct = 1)
Digital Engagement	Online financial activities (qp8 variables)	Average
Financial Capability	Knowledge + self-assessed + digital engagement	Average
Overconfidence Gap	Self-assessed – actual knowledge	Difference

Table B2. Coding Rules

Variable	Transformation
Survival capacity	Reversed: 6 – value
Knowledge variables	Correct = 1, Incorrect = 0
Self-assessed knowledge	Scaled between 0 and 1
Missing values	-97, -98, -99 → NA

Appendix C: Data Preparation Process

Table C1. Dataset Reduction

Step	Variables	Observations
Raw dataset	219	4,862
Selected variables	28	4,862
Final cleaned dataset	22	4,862

Table C2. Data Cleaning Steps

Step	Description
Step 1	Renaming variables using codebook
Step 2	Selecting relevant variables
Step 3	Converting special codes to NA
Step 4	Removing high-missing variables (qk7)
Step 5	Constructing indices

Appendix D: Regression Models

Table D1. Main Regression Model

Model / variable	Coefficient	p-value	Interpretation
Main model: Financial capability	0.0386	0.607	Not significant; capability does not significantly reduce vulnerability
Main model: Income	-0.1206	<0.001	Higher income is associated with lower vulnerability
Main model: Age	-0.0038	<0.001	Older respondents are slightly less vulnerable in this model
Main model: Gender	-0.0260	0.283	Not statistically significant
Model R-squared	0.0238	--	Small explanatory power, expected in social survey data

Table D2. Interaction Model

Variable	Coefficient	p-value	Interpretation
Financial Capability	0.0386	0.607	Not significant
Income	-0.1206	<0.001	Significant negative effect
Capability × Income	~0.000	0.717	No interaction effect
R-squared	~0.024	--	No improvement from interaction

Table D3. Overconfidence Model

Variable	Coefficient	p-value	Interpretation
Overconfidence Gap	0.1323	<0.001	Higher mismatch increases vulnerability
Income	Negative	<0.001	Remains protective
Age	Negative	<0.001	Slight protective effect
Gender	Not significant	>0.05	No effect
R-squared	~0.03	--	Slight improvement

Appendix E: Reproducibility

The analysis was conducted in R using tidyverse and ggplot2.

The full code is available at: